THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The strength of an electromagnet can be increased by 1

(a) increasing the current or the number of turns
(b) decreasing the current
(c) using a thicker iron-free coil
(d) removing the core

2. The direction of the induced current is given by 1

(a) Joule's law **(b)** Fleming's right-hand rule
(c) Fleming's left-hand rule **(d)** Ohm's law

3. A compass needle is deflected when placed near a current-carrying wire. This was observed by 1

(a) Oersted **(b)** Fleming **(c)** Ampere **(d)** Faraday

4. A current-carrying conductor placed in a magnetic field experiences 1

(a) no force **(b)** a decrease in current
(c) a force **(d)** only heating

5. An electric generator converts 1

(a) electrical energy into mechanical energy **(b)** mechanical energy into electrical energy
(c) heat into light **(d)** chemical energy into electrical energy

6. The magnetic field produced by a straight current-carrying wire has field lines in the form of 1

(a) radial lines **(b)** concentric circles around the wire
(c) straight lines parallel to the wire **(d)** ellipses

7. The force on a current-carrying conductor is maximum when the current is 1

(a) parallel to the field **(b)** perpendicular to the magnetic field
(c) at 30 degrees **(d)** zero

8. In a generator, the armature coil rotates in 1

(a) a gravitational field only **(b)** an electric field
(c) vacuum **(d)** a magnetic field

9. The direction of the magnetic field around a straight wire is given by 1

(a) Lenz's rule **(b)** the right-hand thumb rule
(c) Fleming's left-hand rule **(d)** Fleming's right-hand rule

10. The direction of force on a current-carrying conductor is given by 1

(a) Fleming's left-hand rule **(b)** Snell's law

(c) Ohm's law **(d)** Fleming's right-hand rule

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is electromagnetic induction? 2

12. What is a short circuit? 2

13. What is an electromagnet? Give one use. 2

14. State Fleming's right-hand rule. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. How does the magnetic field due to a straight wire change when 3

(a) the current is increased

(b) the distance from the wire is increased?

16. State the difference between an electric generator and the process of electromagnetic induction. 3

17. Why is the metal body of an appliance earthed? 3

OR

What is a solenoid? State the nature of its magnetic field. 3

18. Describe an activity to show the electromagnetic induction using a coil and a magnet. 3

OR

How does a magnetic field exert a force on a current-carrying conductor? Name the physical quantities that affect the force. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. State the rule to find the direction of force on a current-carrying conductor. A conductor is placed in a field directed vertically downwards and the current flows from east to west. Find the direction of the force. 5

OR

Explain why the magnetic field lines around a bar magnet never cross, and how the field strength is shown by them. Mention three properties. 5

20. Describe Faraday's experiments showing electromagnetic induction. State the conditions for the induced current. 5

OR

Describe the right-hand rule for a solenoid and explain the factors affecting the strength of the magnetic field. 5

END OF QUESTION PAPER Total: 40 Marks

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